

RESEARCH ARTICLE

Evaluating Methods for Transplanting Endangered Elkhorn Corals in the Virgin Islands

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Abstract

Restoration of rare corals is desirable and restoration projects are fairly common, but scientific evaluation of this approach is limited. We tested several techniques for transplant and restabilization of *Acropora palmata* (the elkhorn coral), an ecologically important Caribbean coral whose populations have suffered severe declines. In rough weather, fragments break-off colonies of branching corals like *A. palmata* as a normal form of asexual reproduction. We transplanted naturally produced coral fragments from remnant populations to nearby restoration sites. Untouched control fragments at the donor site died faster and grew slower than fragments attached to the reef, so attaching fragments to the reef is beneficial. Transplanted fragments grew and died at a rate similar to fragments left at the donor site (both groups were attached to the reef), so there were no effects of moving fragments or differences in habitat quality between

donor and restoration sites. Growth and survival were similar using four methods of attaching fragments to the reef: cable ties, two types of epoxy resin, and hydrostatic cement. Corals sometimes compete with the macroalgae that dominate degraded reefs, and clearing surrounding algae improved the growth of fragments. After 4 years, transplanted fragments grew to 1,450 cm² in area and so were potentially sexually active. Because the methods tested are simple and cheap, they could be used by volunteer recreational divers to restore locally extirpated *A. palmata* populations or to enhance reefs where natural recovery is slow.

Key words: *Acropora palmata*, Caribbean coral reefs, elkhorn coral, endangered species, grassroots restoration, growth, reef rehabilitation, survival, transplanting methodology.

Introduction

Coral reefs are excellent candidates for ecological restoration because they are in decline globally due to overharvesting, disease outbreaks, climate change, and other localized impacts (Gardner et al. 2003; Bellwood et al. 2004). Historically, restoration has not been a major part of coral reef management and conservation, but it has grown in popularity

over the past decade (Yap 2000; Rinkevich 2005; Precht 2006). Restoration by transplantation is common for terrestrial plants (Menges 2008) and has been used for some marine organisms like seaweeds (Carney et al. 2005), seagrasses (Bastyan & Cambridge 2008), and mangroves (Ellison 2000). Most coral restoration projects have used directly transplanted corals, or coral pieces, from sites with remnant populations, but “garden-ing”—the culture of small coral pieces in nurseries for later transplant—has also been gaining in popularity (Epstein et al. 2003). Restoration projects have used whole coral colonies, seeding of planula larvae, transplanting of tiny “nubbins,” and transplanting of branches or fragments. Various methods of securing corals to the substratum have been used, and trials have been done to identify substrates best suited for attachment. Although coral transplanting has promise and is commonly advocated (Yap 2000; Rinkevich 2005; Abelson 2006; Precht 2006), there are also some potential drawbacks of this approach. Transplanted corals sometimes exhibit poor growth and survival. Direct transplantation often involves loss of coral from the donor site because source corals are broken and/or removed for restoration. Gardening is potentially less damaging to source populations, but is time and labor

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intensive (Harriott & Fisk 1988; Edwards & Clark 1998). Rigorous assessment of these pros and cons of transplanting is, however, difficult because many projects lack appropriate controls against which the performance of transplanted fragments can be compared (Bowden-Kerby 2001; Bruckner & Bruckner 2001), and because there are few direct comparisons of alternate methods of securing fragments to the reef (Bruckner & Bruckner 2001).

We focus on *Acropora palmata* (the elkhorn coral), a major reef-building coral in the Caribbean that was formerly the dominant species in shallow wave-exposed areas. In the past, *A. palmata* ameliorated wave damage along coastlines and provided habitat for fishes and other taxa. During the 1980s and 1990s, *A. palmata* declined in abundance by 85–98% (Gardner et al. 2003) and localized extirpations have occurred. Most mortality is attributed to white-band disease, compounded by storm damage, coral bleaching, and other human impacts (Aronson & Precht 2001; Bowden-Kerby 2001). This decline has no precedent in the last few thousand years (Aronson & Precht 1997, 2001) and prompted listing of *A. palmata* under the U.S. Endangered Species Act (Precht et al. 2004; Federal Register 2006). Any restoration method that would accelerate the natural recovery of *A. palmata* would thus be valuable.

One possibility for restoration of branching corals like *A. palmata* arises from the fact that most reproduction occurs by asexual fragmentation. Fragments break-off colonies during storms and are sometimes able to reattach to the substratum to form new colonies (Bak & Engel 1979; Dunne & Brown 1980; Highsmith et al. 1980; Fong & Lirman 1995). Fragmentation is a life-history trait shared by many species of branching corals (Highsmith 1982; Wallace 1985). At sites where natural recovery has begun by the successful recruitment of planktonic larvae, one obvious question is—does transplanting corals to the site speed recovery? This question was addressed in a prior study by comparing transplanted and naturally recruited corals at a restoration site (Garrison & Ward 2008).

We focused on a different question—when storm-generated fragments appear, is it beneficial to use them for restoration? We considered untouched fragments left at the donor sites as the appropriate control group against which to test the performance of transplanted fragments. We transplanted other fragments from donor sites to nearby restoration sites where *A. palmata* had been extirpated. We isolated the effects on fragment growth and survival of four different components of the restoration protocol. Because we assumed that being attached to the substratum is important for fragment survival, we tested (1) whether securing fragments to the substratum affects their growth and survival. Because handling and transporting fragments is potentially stressful, we also tested (2) whether moving corals between sites affects their growth and survival. Because few methods of securing fragments have directly been compared, we also tested (3) whether fragment growth and survival differed between four simple low-cost methods of attachment. Because macroalgae can compete with corals (McCook et al. 2001), we tested (4) whether scraping algae from around the transplants improved their ability to establish.

Methods

Study Sites

All experiments were done near Guana Island in the British Virgin Islands (18°29'N, 64°35'W). The restoration sites were four shallow (0.4–1.6 m deep) patch reefs on the leeward south side of the island. Based on the presence of skeletal remains and anecdotal reports, *Acropora palmata* was once common on these reefs. Based on a careful search and a survey in 2005 (36 point-intercept transects, each 30 m long), no *A. palmata* colonies remained by 2005 and other coral species were rare (1.6% live cover). The substratum is limestone pavement and rubble, overlain primarily with macroalgae and zooanthids (60 and 16% cover, respectively in 2005). We used two sites as sources of *A. palmata* fragments, one on the leeward south side of the island 0.4 km from the transplant sites and a more exposed site on the eastern side of the island 3.6 km away. Both source sites were steep rock walls close to the island and fragments were taken from 1.6 to 6.4 m.

Study Design

We conducted two main experiments: experiments 1 and 2, which started in 2007 and 2008, respectively. Experiment 1 was done on the patch reef furthest west in the study area, and experiment 2 on the reef immediately east. Both experiments used an asymmetrical analysis of variance design (Underwood 1997) to isolate two key steps in restoration: stabilizing fragments by attaching them to the reef and relocating them to the restoration site (Fig. 1). Isolating these steps required three treatments: (1) control fragments left unattached at the source sites; (2) fragments attached to the reef at the source sites; and (3) fragments that were moved to the restoration site and attached (Fig. 1). Comparing treatments (1) and (2) isolated the effect of attaching fragments, and comparing (2) with (3) isolated the effect of relocating fragments.

Another goal of the experiments was to compare the methods of attaching fragments to the reef. In experiment 1, we compared two methods: Z-spar epoxy resin and cable ties. In experiment 2, we compared four methods: Z-spar epoxy, cable ties, PC Marine Epoxy Putty, and hydrostatic cement. For each method of attachment, roughly 50% of fragments were attached at the source site and 50% at the restoration site (Fig. 1).

Our final aim was to test whether scraping macroalgae from around the secured fragments improved their growth and survival. This factor was tested only in experiment 2. The “algal removal” treatment was orthogonal with the “transplant” and “method of attachment” treatments, that is scraped and unscraped fragments were divided equally among the other treatments (Fig. 1).

We also conducted a preliminary experiment in 2005, in which control fragments were compared to a group that was moved to four of the restoration patch reefs and fixed to the substratum with Z-spar epoxy (Fig. 1). We compared these two treatments to assess the overall restoration protocol over a longer period (4 years).

(a) 2005 experiment

moved?	secured?	how secured?	(n)
no	no		(7)
yes	yes	epoxy1	(35)

(a) 2007 experiment

moved?	secured?	how secured?	(n)
no	no		(29)
no	yes	epoxy1	(13)
		cable tie	(14)
yes	yes	epoxy1	(15)
		cable tie	(15)

(b) 2008 experiment

moved?	secured?	how secured?	algae scraped?	(n)
no	no			(43)
		epoxy1	y	(4)
			n	(4)
no	yes	cable tie	y	(11)
			n	(11)
		epoxy2	y	(9)
			n	(9)
		cement	y	(11)
			n	(11)
		epoxy1	y	(25)
			n	(25)
		cable tie	y	(39)
			n	(42)
yes	yes	epoxy2	y	(9)
			n	(9)
		cement	y	(9)
			n	(9)

Figure 1. Summary of the experimental design, indicating whether corals were transplanted (moved), attached to the reef (secured), and the method of attachment (how secured). Epoxy 1 is Z-spar and epoxy 2 is PC Marine epoxy putty. For algal scraping, y = yes, n = no. Sample size for each treatment is indicated (n).

Experimental Procedures

All three experiments started between late July and mid-August (except for 25 fragments in 2005 that were transplanted in November). Control fragments were untouched, but marked by securing a numbered plastic tag to the reef nearby. Fragments to be transplanted were removed by SCUBA divers. Divers wore rubber gloves and placed each fragment in a plastic bag, so that fragments would not touch each other during transport. Bagged fragments were carried to the surface and placed in plastic bins of seawater. Bins were covered and seawater was changed regularly to minimize warming, and corals were transplanted within 2.5 hours of being removed.

Before attaching a coral to the reef, we scraped and brushed the attachment site and secured a numbered plastic tag to the reef nearby. Fragments attached using cable ties were tied to small projections on the reef or to branches of dead *A. palmata* using 1–3 ties. We wrapped the cable ties around dead areas of the fragment when possible. The two types of

epoxy resin used were Z-spar A788 splash zone compound (Carboline Company, St. Louis, MO, U.S.A.) and PC Marine Epoxy Putty (PC-Products, Allentown, PA, U.S.A.). The two components of Z-spar were mixed on the boat and placed in a plastic bag for use underwater, whereas PC Marine Epoxy was packaged as a stick and was mixed underwater. In both cases, the putty-like resin was molded around the base of the fragment to secure it to the reef. The cement mixture comprised Portland cement (1,200 mL), hydrostatic cement (400 mL), sand (200 mL), warm seawater (600 mL), and antiplumbing agent (7 mL). These ingredients were mixed in a bucket to form a stiff paste, and the bucket plus mixture was taken underwater by divers. Cement was pressed to the cleaned reef surface by hand, and then molded around the base of a coral fragment.

In experiment 2, to test whether removing macroalgae improved fragment growth and survival, we spent 2–3 minutes scraping macroalgae, primarily *Dictyota* spp., *Dictyosphaeria cavernosa* ([Forsskål] Børgesen) and *Lobophora variegata* ([J.V. Lamouroux] Womersley ex Oliveira) from a circle of 20 cm radius around each fragment.

Measuring Coral Growth and Survival

We used the surface area of live tissue (leaf area index [LAI]) as an index of fragment size, because *A. palmata* has a branching morphology and sometimes suffers partial colony mortality, as well as regrowth over the skeleton by new tissue. In experiments 1 and 2, we estimated LAI by taking several digital photographs of each coral, with a ruler in the frame to provide a scale (Bythell et al. 2001). We photographed all major flat surfaces and used image analysis software (ImageJ) to trace the perimeter of each surface and calculate their areas (Abramoff et al. 2004). The status of these fragments was checked after 2 months and 1 year, when we noted which fragments had died and remeasured those with live tissue. As an index of growth or shrinkage, we calculated the % change in LAI. In the 2005 experiment, we estimated LAI using a ruler as $([L + W + H]/3)^2$, where L = length, W = width, and H = height in cm of live tissue (Williams et al. 2008). We assessed the status of these fragments annually for 4 years. The fragments used in the three experiments ranged in initial LAI from 7.0 to 961 cm² in 2005 (mean $\pm$ SD = 161 $\pm$ 192), from 8.5 to 950 cm² in 2007 (mean $\pm$ SD = 117.8 $\pm$ 148.2), and from 5.2 to 822.0 cm² in 2008 (mean $\pm$ SD = 106.4 $\pm$ 113.3).

Data Analysis

For experiments 1 and 2, we performed a series of planned comparisons using analysis of covariance (ANCOVA) to test the effects of the treatments on the initial performance of the transplants (Quinn & Keough 2002). The response variable was % change in LAI of all fragments after 2 months, so it provides a composite measure of growth and survival (a fragment that died would be scored –100%). To isolate the effect of securing fragments to the reef, we first considered only

fragments at the donor site and compared control fragments to all secured fragments. To isolate the effect of moving fragments, we next considered only fragments that were secured to the reef, and compared fragments that were moved to those not. The third analysis compared the methods of securing fragments to the reef. For experiment 2, we performed a fourth test comparing fragments cleared of surrounding algae to equivalent fragments that were not cleared. Initial fragment size was included as a covariate in each ANCOVA to control for the weak negative relationship between initial fragment size and growth. LAI data were arc-sine transformed before analysis, after which they met the assumptions of ANCOVA. For experiments 1 and 2, we also tested treatment effects on growth and survival over the first year. The number of fragments that survived was compared among treatments using chi-square tests. Treatment effects on growth were tested using ANCOVA (as described earlier), but the response variable was % change in LAI of surviving fragments.

Results

Does Securing Fragments to the Substratum Affect Their Growth and Survival?

Over the first 2 months, unattached fragments noticeably decreased in LAI due the partial or complete mortality of many fragments, whereas fragments secured to the reef did significantly better (Fig. 2). This was true in both experiments 1 and 2 (experiment 1 ANCOVA: $F_{[1,53]} = 5.49$, $p = 0.023$; experiment 2 ANCOVA: $F_{[1,96]} = 10.62$, $p = 0.002$). After 1 year, very few of the unattached fragments remained alive (3 and 12% in experiments 1 and 2, respectively), but roughly half of their attached counterparts survived (56 and 62% of those in experiments 1 and 2, respectively). These differences in the number of survivors were statistically significant in both experiments (experiment 1 $\chi^2 = 13.3$, $df = 1$, $p = 0.0003$; experiment 2 $\chi^2 = 15.3$, $df = 1$, $p < 0.0001$). Because so few unattached corals survived, we did not compare their growth to the attached fragments. In summary, securing fragments to the substratum caused increases in initial tissue growth and subsequent survival.

Does Moving Corals between Sites Affect Their Growth and Survival?

Isolating the effect of moving fragments showed no influence of this treatment on % change in LAI after 2 months. This was true in experiment 1, when most fragments initially gained live tissue (ANCOVA: $F_{[1,54]} = 0.001$, $p = 0.971$), and in experiment 2 when fragments initially decreased in LAI (ANCOVA: $F_{[1,200]} = 0.046$, $p = 0.831$) (Fig. 3).

After a year, fragments at the source site survived about as well (56 and 62% surviving in experiments 1 and 2, respectively) as those moved to the restoration site (30 and 54% in experiments 1 and 2, respectively), and relocation had no detectable influence on survival (experiment 1: $\chi^2 = 2.2$, $df = 1$, $p = 0.138$; experiment 2: $\chi^2 = 0.42$, $df = 1$,

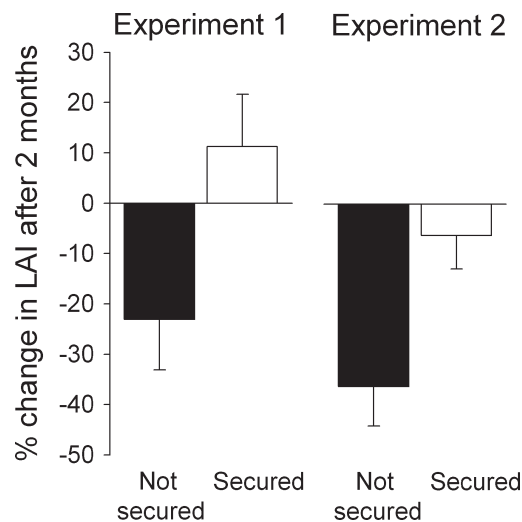


Figure 2. Effects of securing fragments to the substratum on % change in area of live tissue (LAI) after 2 months. Plotted are means $\pm$ SE. Data include all fragments, so fragments that died would be scored -100% .

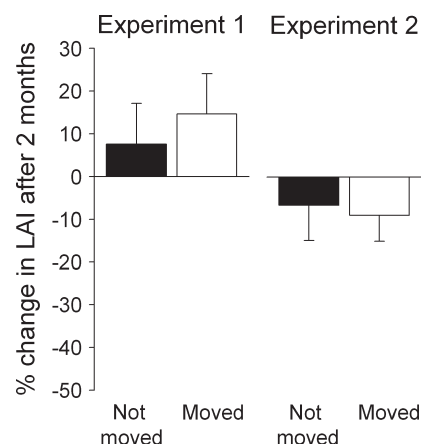


Figure 3. Effects of moving fragments to a restoration site on % change in area of live tissue (LAI) after 2 months. Plotted are means $\pm$ SE. Data are presented for all fragments, so fragments that died would be scored -100% .

$p < 0.516$). Fragments that survived the first year all grew substantially (Fig. 4), and there was no detectable difference in growth between those that were moved and those that remained at the donor site (experiment 1 ANCOVA: $F_{[1,19]} = 0.855$, $p = 0.375$; experiment 2 ANCOVA: $F_{[1,114]} = 1.64$, $p = 0.297$). Overall, the process of moving fragments appears to have no marked influence on tissue growth or survival.

Does the Method of Securing Fragments Affect Their Growth and Survival?

In experiment 1, when we compared fragments secured with cable ties and Z-spar epoxy, fragments attached using both methods accrued live tissue over the first 2 months (Fig. 5), and attachment method had no significant effect on LAI (ANCOVA: $F_{[1,54]} = 0.581$, $p = 0.449$). The method used to attach fragments also had no discernible effect on how many

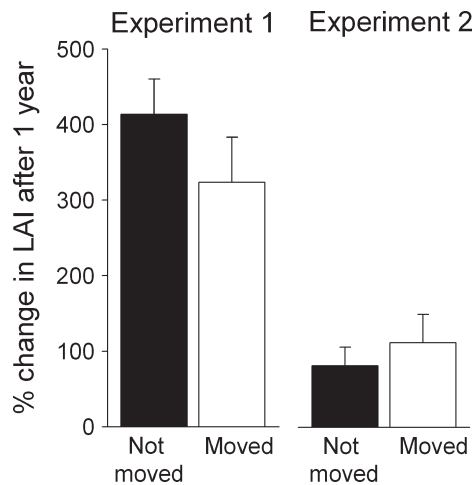


Figure 4. Effects of moving fragments to a restoration site on % change in area of live tissue (LAI) after 1 year. Plotted are means $\pm$ SE. Data include only fragments that survived.

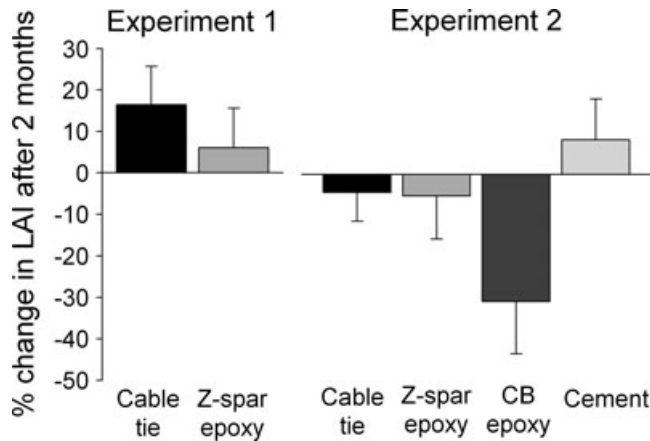


Figure 5. Effects of four methods of attaching fragments to the reef on % change in area of live tissue (LAI) after 2 months. Fragments were secured using cable ties, hydrostatic cement, and two types of marine epoxy. Plotted are means $\pm$ SE. Data are presented for all fragments, so fragments that died would be scored -100%.

fragments survived their first year ($\chi^2 = 0.145$, $df = 1$, $p = 0.702$), nor on the growth of surviving fragments (ANCOVA: $F_{[2,19]} = 1.237$, $p = 0.327$) (Fig. 6).

When we expanded our design in experiment 2 to compare the four methods of attachment, there were also no statistically significant differences in % change of LAI over 2 months due to the method of attachment (ANCOVA: $F_{[3,200]} = 1.995$, $p = 0.116$). After 1 year, roughly half of the fragments remained alive in all four treatments (% survival by method: Z-spar = 50%, PC marine Epoxy = 44%, cable ties = 56%, cement = 66%) and the number of survivors did not differ among treatments ($\chi^2 = 3.71$, $df = 1$, $p = 0.294$). Similarly, surviving fragments grew at similar rates regardless of how they were attached to the reef (Fig. 6) (ANCOVA: $F_{[3,115]} = 1.99$, $p = 0.119$). Overall, coral fragments appeared to fare

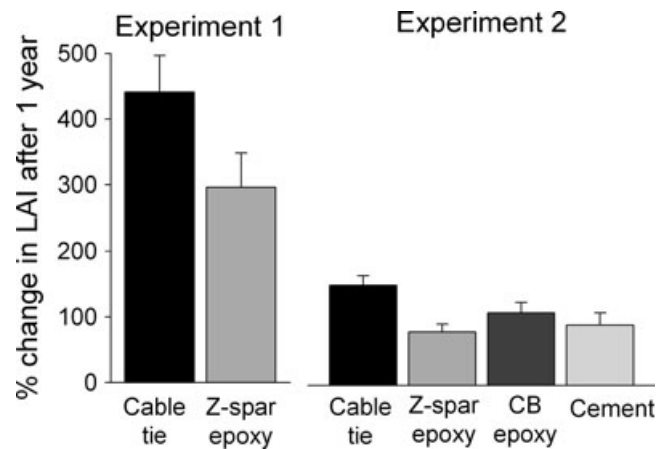


Figure 6. Effects of four methods of attaching fragments to the reef on % change in area of live tissue (LAI) after 1 year. Fragments were secured using cable ties, hydrostatic cement, and two types of marine epoxy. Plotted are means $\pm$ SE. Data include only fragments that survived.

equally well regardless of the method used to attach them to the reef.

Does Clearing Macroalgae From around Fragments Improve Their Growth and Survival?

Over the first 2 months, fragments cleared of surrounding macroalgae showed similar changes in LAI to those still surrounded by algae (ANCOVA: $F_{[1,202]} = 0.143$, $p = 0.705$) (Fig. 7). After 1 year, the survival of fragments was also unaffected by clearing (% survival with algae cleared = 52%, % survival without = 60%; $\chi^2 = 0.665$, $df = 1$, $p = 0.658$). However, surviving fragments grew substantially larger if cleared of surrounding algae than if not (Fig. 7) (ANCOVA: $F_{[1,115]} = 7.34$, $p = 0.007$). Overall, removing macroalgae provides no detectable benefit during the first 2 months, nor any subsequent enhancement of survival, but improved the growth of surviving fragments over 1 year.

Does Transplanting Corals Affect Their Long-Term Growth and Survival?

Although small sample sizes precluded statistical testing, control corals fared poorly in the 2005 experiment. Only 1 of the 7 control fragments from 2005 remained alive after 1 year, and all were dead after 2 years. In contrast, 28 of the 35 transplanted fragments survived the first year and 14 (40%) remained alive after 4 years. Those 14 surviving fragments increased in LAI by 1160% on average, reaching a mean LAI of 1453 cm² (± 462 cm² SD).

Discussion

We isolated the effects of four methodological components of coral transplanting: attachment to the substratum, transporting to a new site, clearing algae around the fragment,

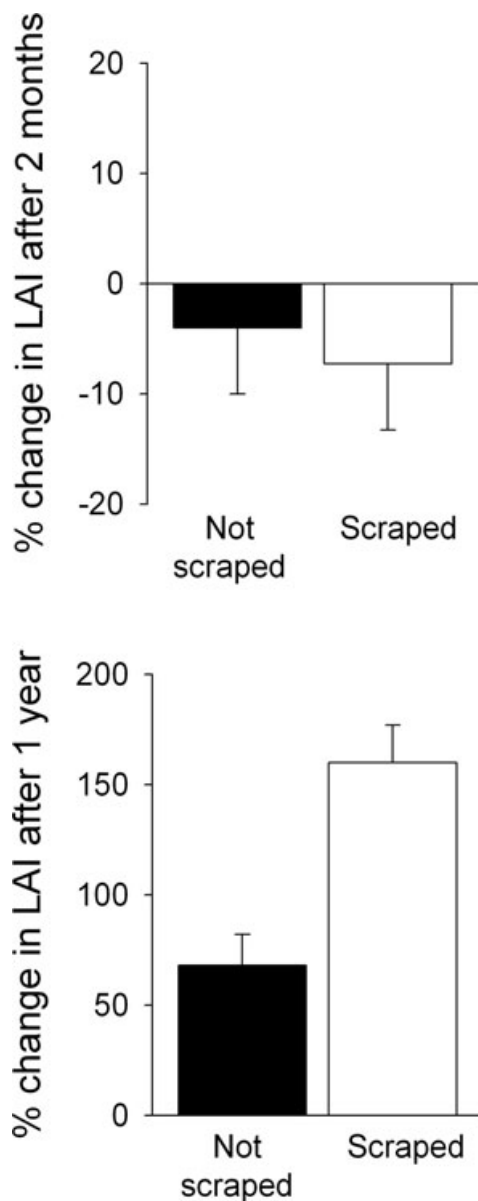


Figure 7. Effects of scraping away surrounding macroalgae on % change in area of live tissue (LAI) after 2 months (upper plot) and 1 year (lower plot). Plotted are means $\pm$ SE. Data for first 2 months included all fragments, so fragments that died would be scored -100% , whereas data for 1 year included only fragments that survived.

and the method of attachment. The first component, attaching fragments to the reef, dramatically improved survival. Unattached fragments of other branching coral species also survive poorly (Bowden-Kerby 2001; Lindahl 2003), apparently because water motion moves loose fragments, which may lead to damage or burial in sediments (Smith & Hughes 1999). In nature, fragments that survive thus tend to be ones that are able to reattach themselves to the substratum (Lirman & Fong 1997; Smith & Hughes 1999) and the survival of reattached fragments (roughly 50% over a year) is within the

range of survival rates reported for natural colonies of similar size (Garrison & Ward 2008).

We were not able to completely isolate the effects of transporting corals, because the effect of transporting is confounded with effects of inherent differences between sites. Site-specific differences in coral growth and survival are common (Plucer-Rosario & Randall 1987; Nagelkerken et al. 2000; Raymundo 2001) due to environmental variation. We anticipated negative site-specific effects because the restoration sites were mostly shallower, so transplanted corals would experience sudden increases in temperature, light and ultraviolet (UV) exposure, and because the restoration site has more sediment in the water. In addition, we anticipated negative impacts of handling and transporting the corals because we took limited precautions when moving them. We kept the protocol simple because we were interested in developing methods usable by volunteer divers with limited training and facilities. The lack of actual site-specific and transplanting effects is thus encouraging, given that the anticipated effects were negative.

Also encouraging is the fact that 40% of the 2005 fragments remained alive after 4 years, despite a major coral bleaching event in Fall 2005, a smaller event in 2006 (Wilkinson & Souter 2008), and a storm swell in March 2008, which damaged shallow reefs throughout the Virgin Islands. More importantly, perhaps, the colonies that remained alive after 4 years were large enough to be reproducing sexually. They could, therefore, potentially begin to release pelagic larvae that could recolonize sites over a wide area (Lirman 2000). The potential of transplant populations to become sources of sexual recruits suggests a high value of future research on the genetic make-up of restored populations. At the simplest level, it will be important to avoid transplanting fragments that are all clones of one genotype (Baums et al. 2005). More complex issues involve the potential for local adaptation and the potential to selectively transplant genotypes with desirable traits, such as disease resistance (Baums 2008).

All four of the attachment methods we compared have been used in restoration projects, as have others we did not test (e.g., tying fragments with steel wire [Bruckner & Bruckner 2001], fishing line [Soong & Chen 2003], or nylon string [Harriott & Fisk 1988]). Most studies in the primary literature used one method (Auberson 1982; Guzman et al. 1991; Yap et al. 1992; Okubo et al. 2005; Garrison & Ward 2008), but a few comparative trials were reported. For example, Dizon and coworkers compared two types of epoxy and superglue (Dizon et al. 2008), whereas Borneman and Lowrie compared epoxy and super glue (Borneman & Lowrie 2001). We found no influence of attachment method on the growth and survival of fragments. Some of the fragments that died in our study appeared to fail because they became loose or detached. This cause of failure was not unique to any one method and has been reported by others for cable ties (Bruckner & Bruckner 2001), epoxy (Borneman & Lowrie 2001), and nylon string (Harriott & Fisk 1988). We observed that firmly attached fragments could grow over their cable ties, and beyond the epoxy or cement holding them down within 2.5 months, so

we hypothesize that any method that keeps the coral firmly attached until overgrowth occurs should be successful. We did note differences among methods of attaching corals in financial cost and ease of use. Cable ties are cheaper and quicker to use than the other methods, so might be preferred where there are enough projections on the reef around which to wrap the ties. At sites where this is not possible, cement might be preferred over epoxy because it is less expensive.

Macroalgae often become dominant on degraded reefs and can outcompete corals, although the outcome of competition is species and site specific (Jompa & Mccook 2003; Nugues & Bak 2006; Smith et al. 2006). Our hypothesis that fragments might benefit from removal of surrounding algae was not supported over the first 2 months, possibly because a large storm in March 2008 reduced macroalgal cover at the restoration site from 60 to 18%. Subsequently, algal cover increased and competitive effects were strong enough to reduce fragment growth by more than half. Our results thus provide the first evidence that clearing algal competitors improves the performance of transplanted corals.

Conclusion

Since our study was done at a few sites over a few years, the generality of the findings is limited. Nonetheless, our results add support to the argument that, when coral colonies are broken and fragmented, either naturally by storms or by human activities such as diving, anchoring, and boat groundings, using these fragments for restoration is beneficial. The direct transplanting of coral fragments is labor intensive enough that it can only be used for modest localized restoration efforts. Although this limits its value, given the large scale over which many human impacts on coral reefs operate (Edwards & Gomez 2007), the simplicity of this approach makes it suitable for volunteers with little training (Bowden-Kerby 2001). These coral transplanting methods can thus be woven into volunteer and educational programs as a way of coupling practical conservation with increased public awareness.

Implications for Practice

- Storm-generated fragments of elkhorn coral used for restoration grow and survive better than fragments left untouched, so intervening does no harm and is likely beneficial.
- Cable ties, epoxy, and cement can all be used to attach coral fragments as long as the fragment is attached securely so it can overgrow the attachment point.
- Cable ties are cheap and quick to use where fragments can be attached to projections on the reef, whereas cement and epoxy might be favored at low-relief sites.
- Moving and reattaching elkhorn coral fragments can be done by recreational divers with minimal training, so can be woven into public educational activities and adopted by volunteer programs.

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